# Appendix

## Appendix A – Exploratory Data Analysis (EDA)

### 1. Cleaned & Documented Dataset

| | |
|---|---|
| Source | AB_NYC_2019.csv |
| Raw rows | 48,895 |
| Removed | 11 zero-price entries (data errors) |
| Final rows | 48,884 \| 20 columns \| 0 missing values |

### 2. Summary of Preprocessing Decisions

**Zero-price listings removed**
11 listings with price = $0 were dropped as data entry errors. All other listings, including luxury outliers up to $10,000, are retained.

**Missing values : reviews_per_month**
10,052 missing values filled with 0. These listings have never been reviewed, so a zero rate is factually correct, not a random gap.

**Missing values : last_review**
Converted to days_since_review (days before 2019-12-31). Missing values filled with the median. New and inactive listings are treated conservatively since we cannot distinguish between them.

**minimum_nights capped at 365**
A small number of listings had values above 365, which is operationally impossible for a short-term rental platform. Capped at 365.

**name and host_name dropped**
Free-text fields with no use in regression or clustering.

### 3. Key EDA Insights

**Location is the dominant price driver.**
Borough accounts for the largest variation in price. Manhattan has the highest median at $150/night; Staten Island the lowest at $62/night. All numerical features have very weak correlations with price ($|r| < 0.1$), confirming that categorical features carry the signal.

**Room type is the second strongest driver.**
Entire homes (52% of listings) command the highest prices. Private rooms (45.7%) sit in the mid-range. Shared rooms (2.4%) form a distinct budget segment.

**The market is geographically concentrated.**

Manhattan and Brooklyn together account for 85.4% of all listings. Predictions for the Bronx (2.2%) and Staten Island (0.8%) will be less reliable due to limited data ( worth noting as a report limitation).

**High-priced listings book less frequently.**
reviews_per_month shows a weak negative correlation with price ($r \approx -0.037$). Higher-priced listings earn more per stay but attract fewer bookings.

**availability_365 = 0 is ambiguous.**
A zero value could mean fully booked, host-blocked, or inactive. **This cannot be resolved from the data alone and should be flagged as a dataset limitation in the report.**

## 4. Selected Features & Rationale for Modelling

Target variable

1. **log_price** : log-transformed price. Normalises the right-skewed distribution so luxury listings do not dominate the model. Convert predictions back to USD with expm1.

Location features

2. **neighbourhood_group** : borough (5 categories, one-hot encoded). Strongest price predictor.
3. **neighbourhood** : 221 unique values, target encoded (replaced by median price per neighbourhood). Avoids 221 dummy columns while retaining pricing signal.
4. **latitude / longitude** : continuous spatial coordinates capturing micro-location effects.

Property features

5. **room_type** : one-hot encoded. Second strongest price predictor.
6. **minimum_nights** : signals short-stay vs long-stay positioning.

Demand & activity features

7. **number_of_reviews** : raw count of guest reviews.
8. **reviews_per_month** : demand velocity; how actively the listing is being booked.
9. **days_since_review** : recency of listing activity (engineered from last_review).
10. **has_reviews** : binary flag: listing maturity proxy. Avoids the skew of raw review counts.

Availability & host features

11. **availability_ratio** : availability_365 / 365. More interpretable as a 0–1 proportion than raw days.
12. **is_multi_host** : binary flag: professional operator (>1 listing) vs casual host. **calculated_host_listings_count** : host portfolio size.

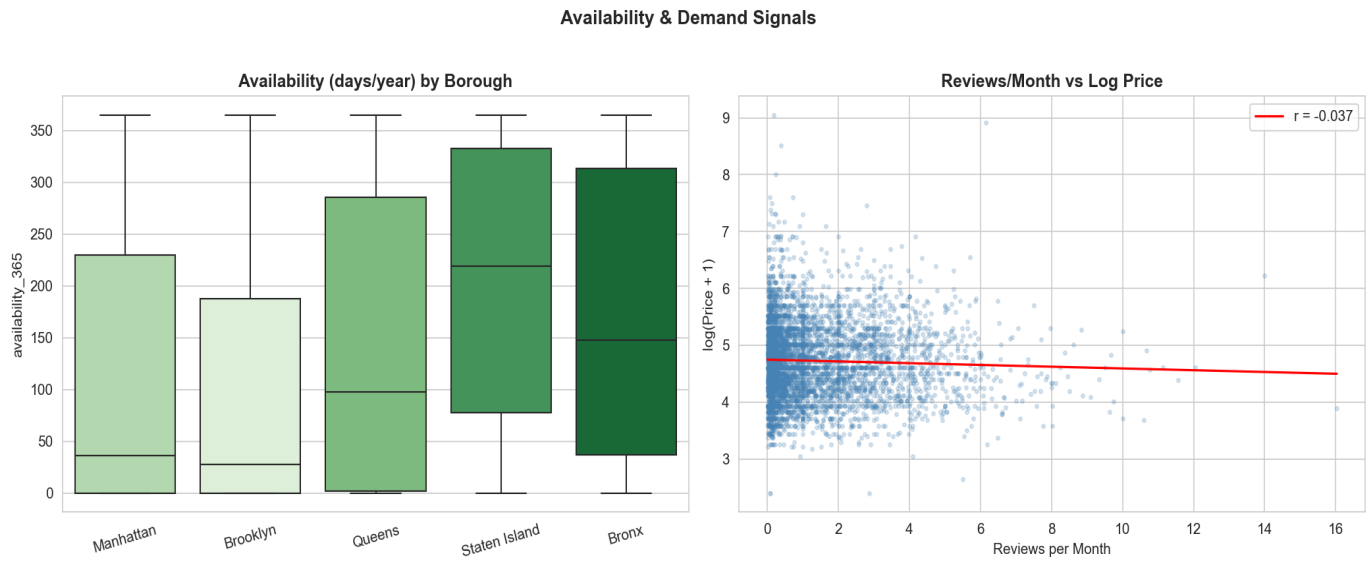


Figure A1: Availability and demand signals across boroughs and listings.

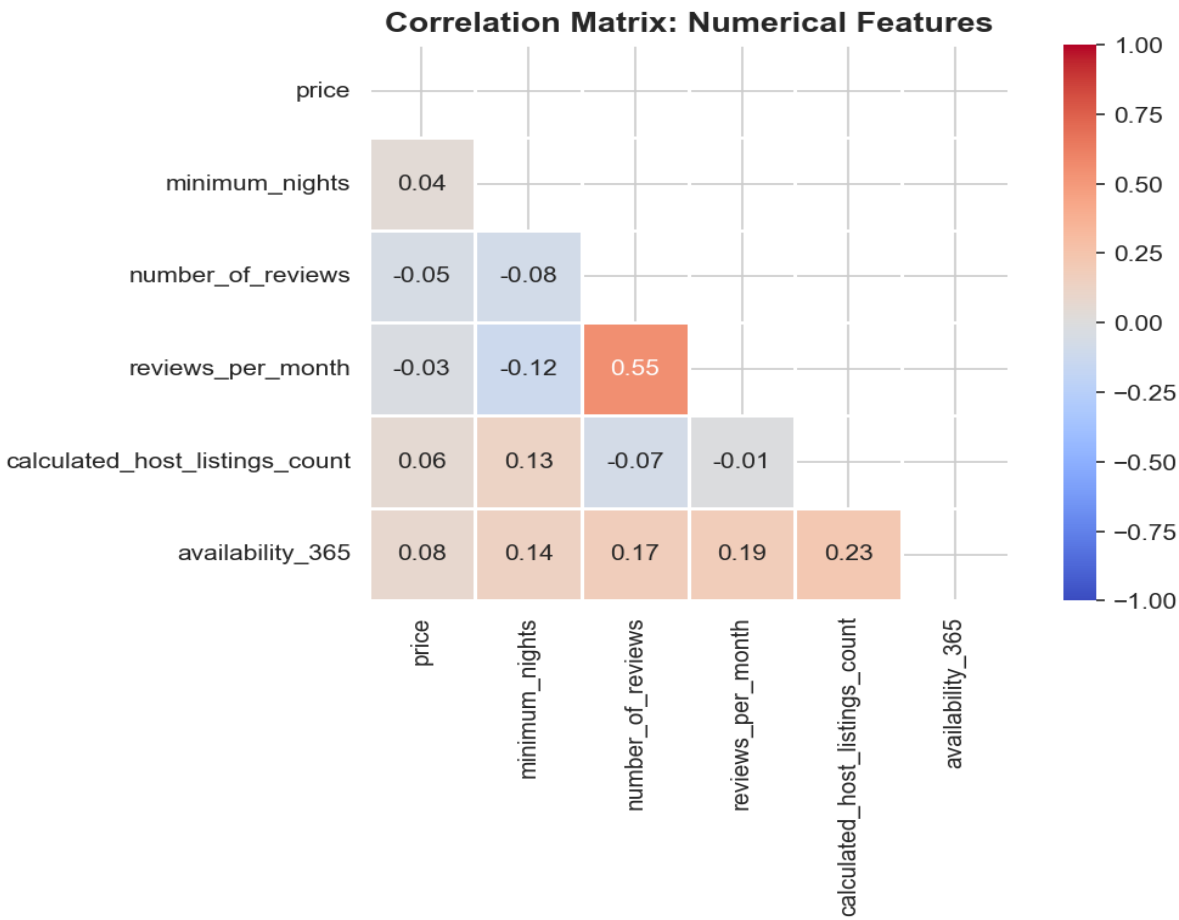


Figure A2: Correlation matrix of key numerical variables in the Airbnb dataset.

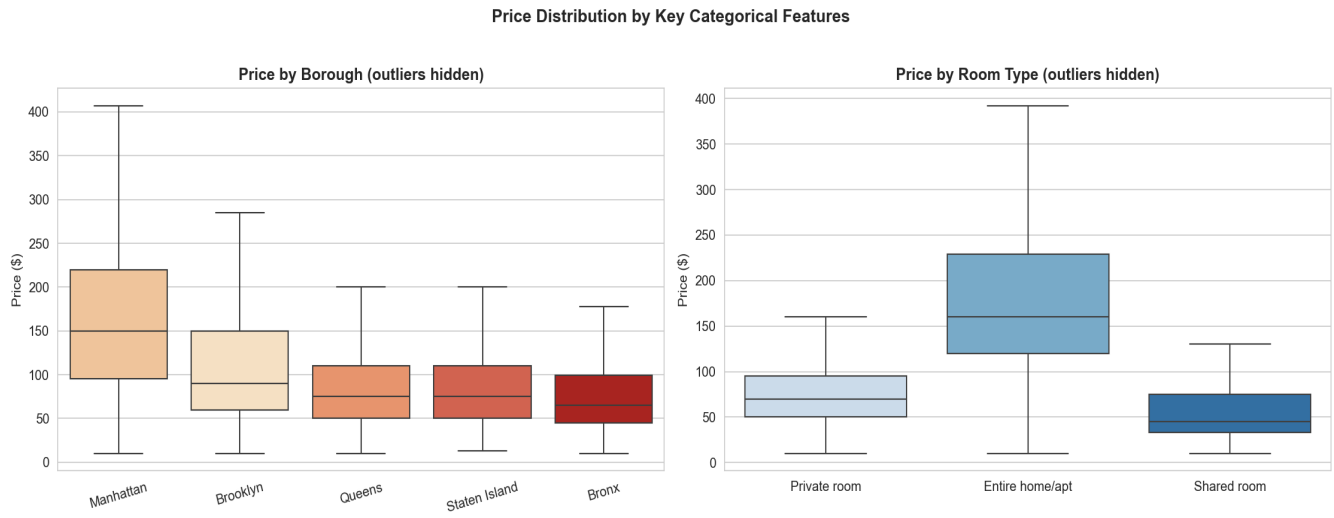


Figure A3: Price distribution across boroughs and room types (outliers hidden).

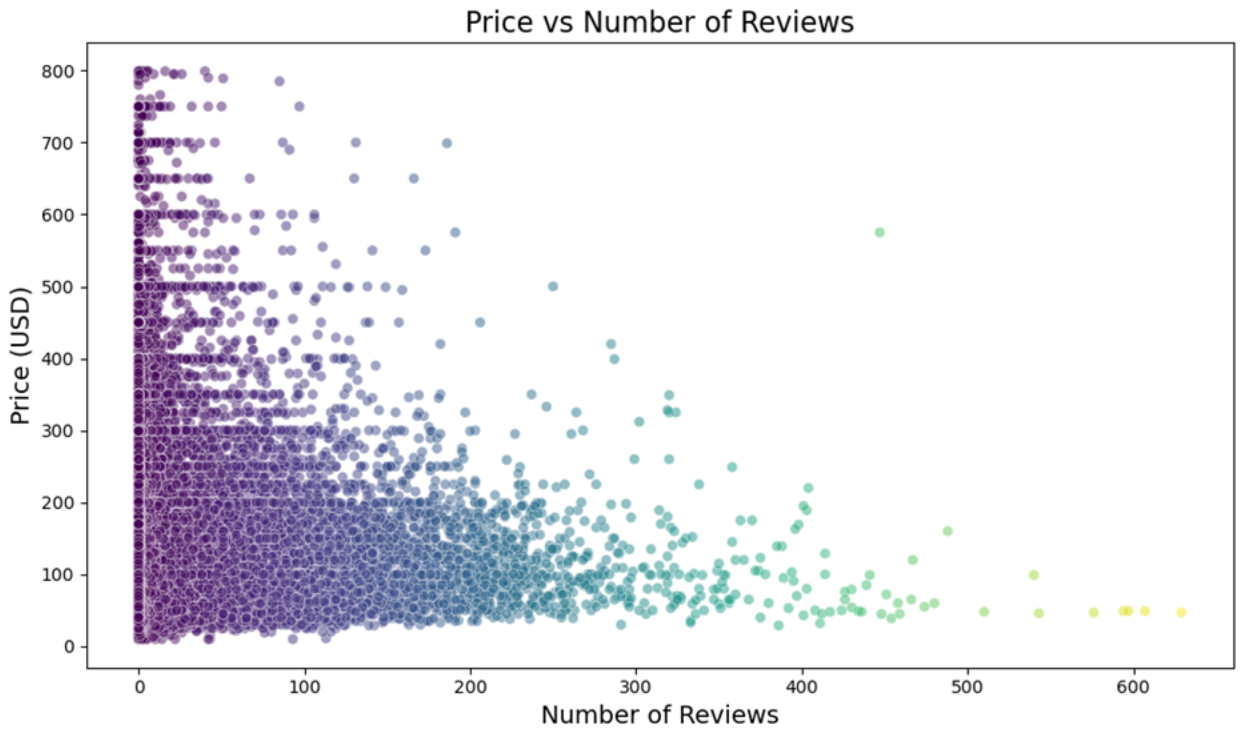


Figure A4: Relationship between listing price and number of reviews.

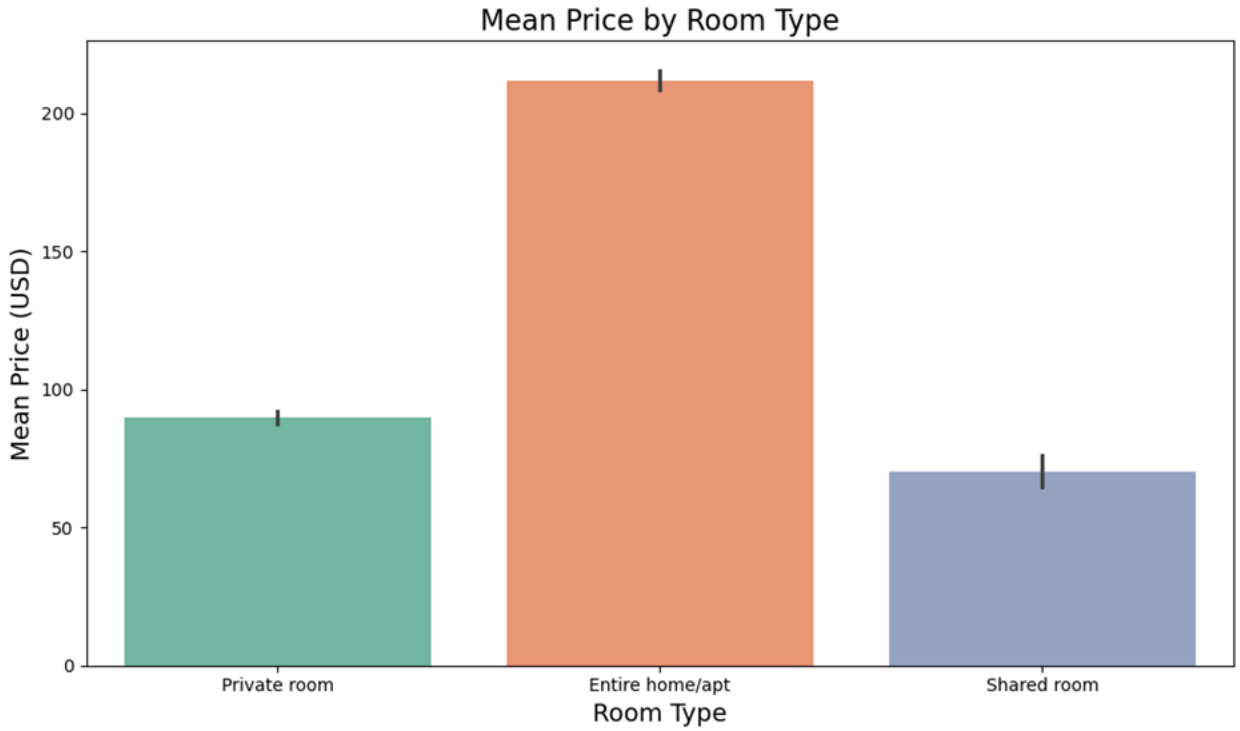


Figure A5: Mean nightly price by room type.

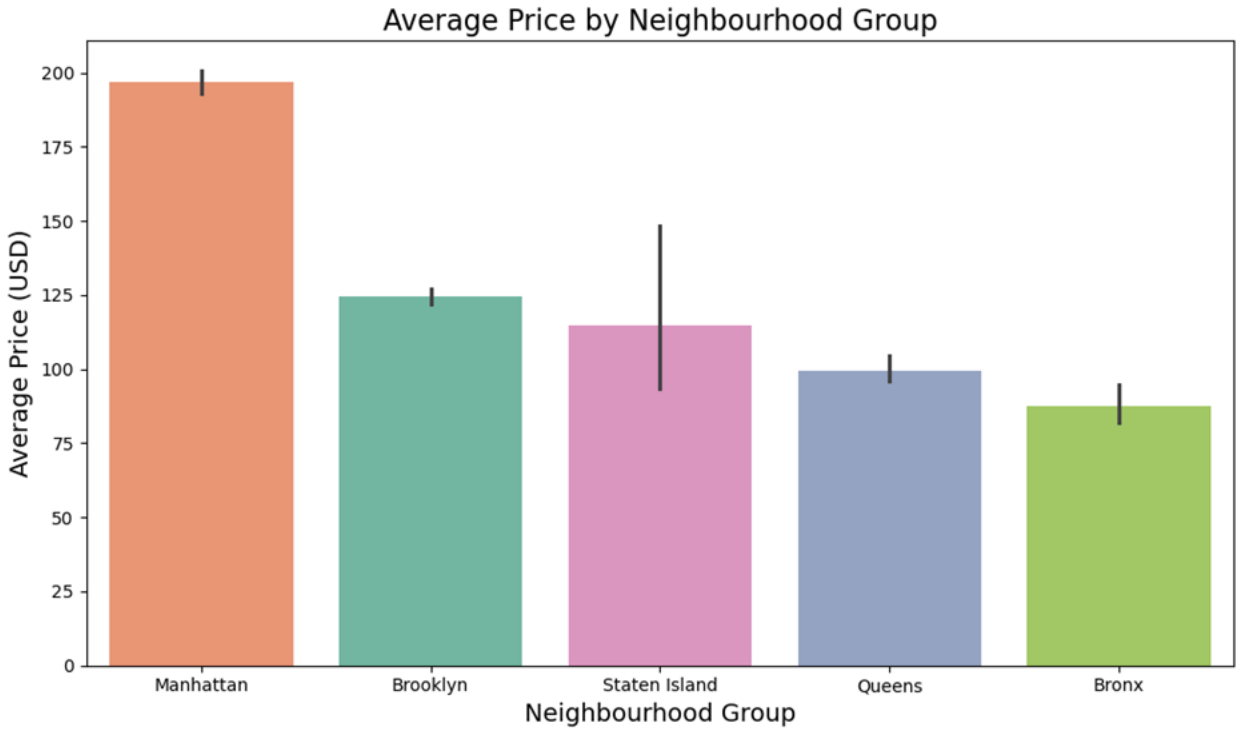


Figure A6: Average nightly price by neighbourhood group in New York City.

Figure A7: Price distribution across the top 10 neighbourhoods in New York City.

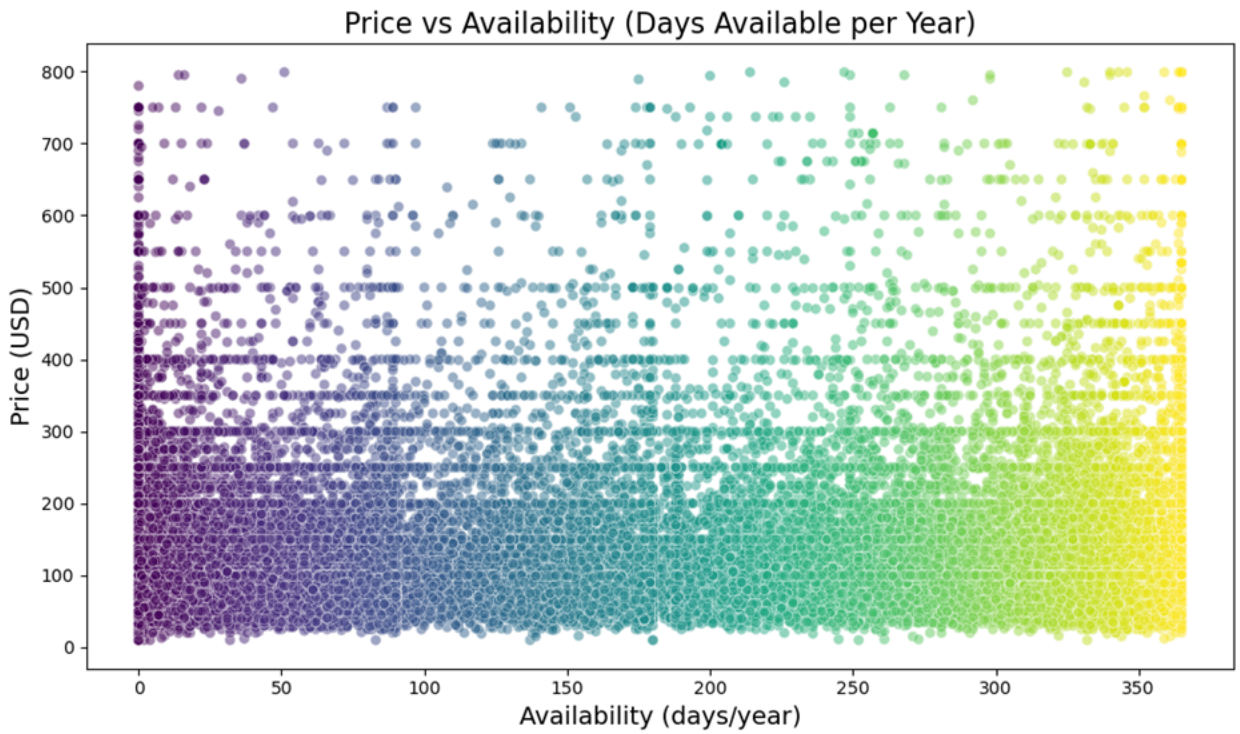


Figure A8: Relationship between listing price and yearly availability (days available per year).

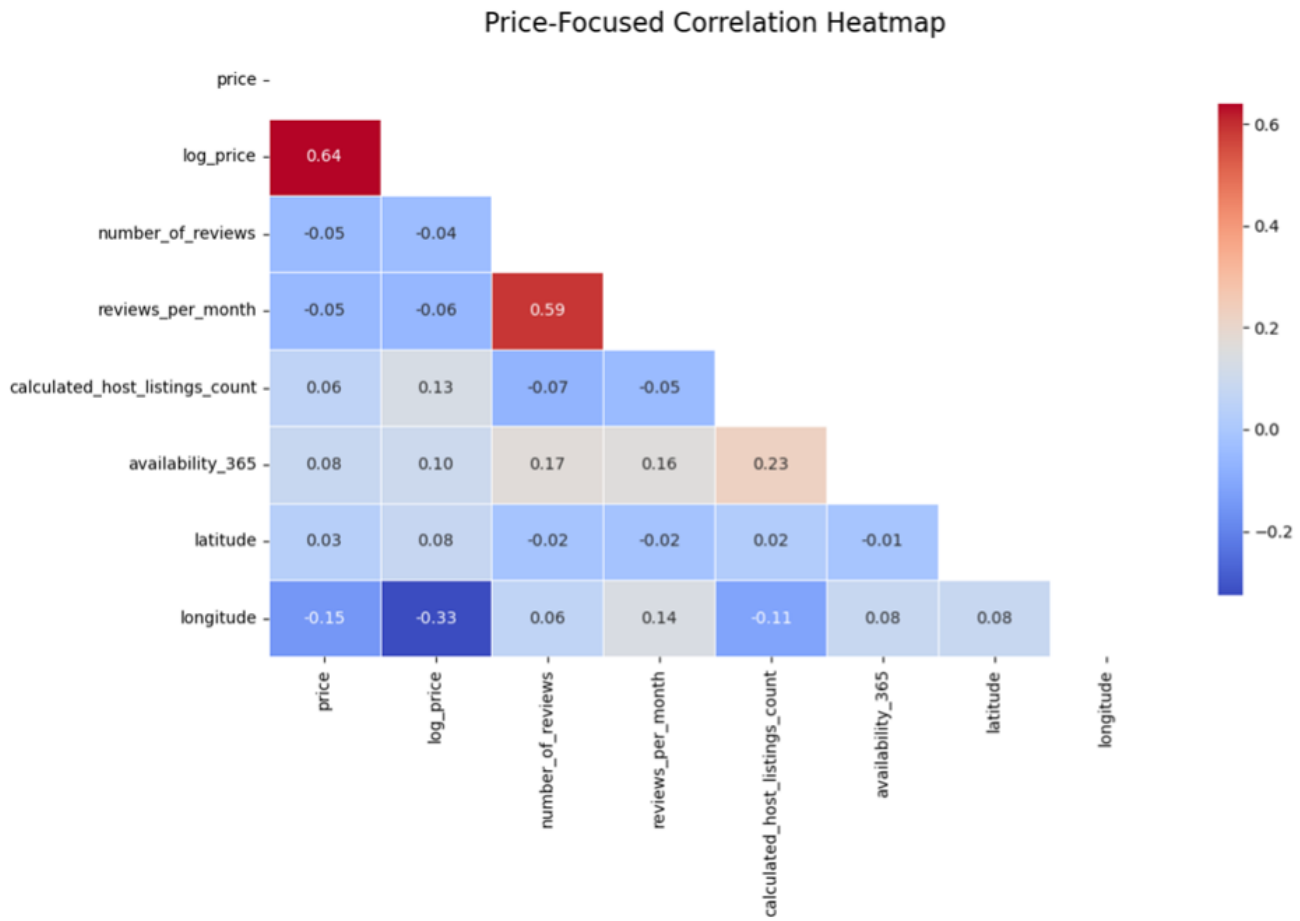


Figure A9: Price-focused correlation heatmap of key numerical features.

## Appendix B – Regression Analysis

**Key Findings from Linear Regression**

- Price is Multivariate

Individual variables showed very low standalone explanatory power ($R^2$ between 0.00–0.02), indicating that pricing is driven by a combination of interacting features.

- Professional Hosts Charge More

Listings managed by hosts with multiple properties (calculated_host_listings_count) show a positive relationship with price, suggesting commercial optimisation behaviour.

- High Review Volume Linked to Lower Prices

Listings with higher numbers of reviews and reviews per month tend to have slightly lower prices, indicating a volume-based competitive strategy.

- Moderate Overall Model Performance

The multivariate regression model achieved approximately $R^2 \sim 0.53$, explaining over half of the variance in listing prices but indicating remaining structural complexity.

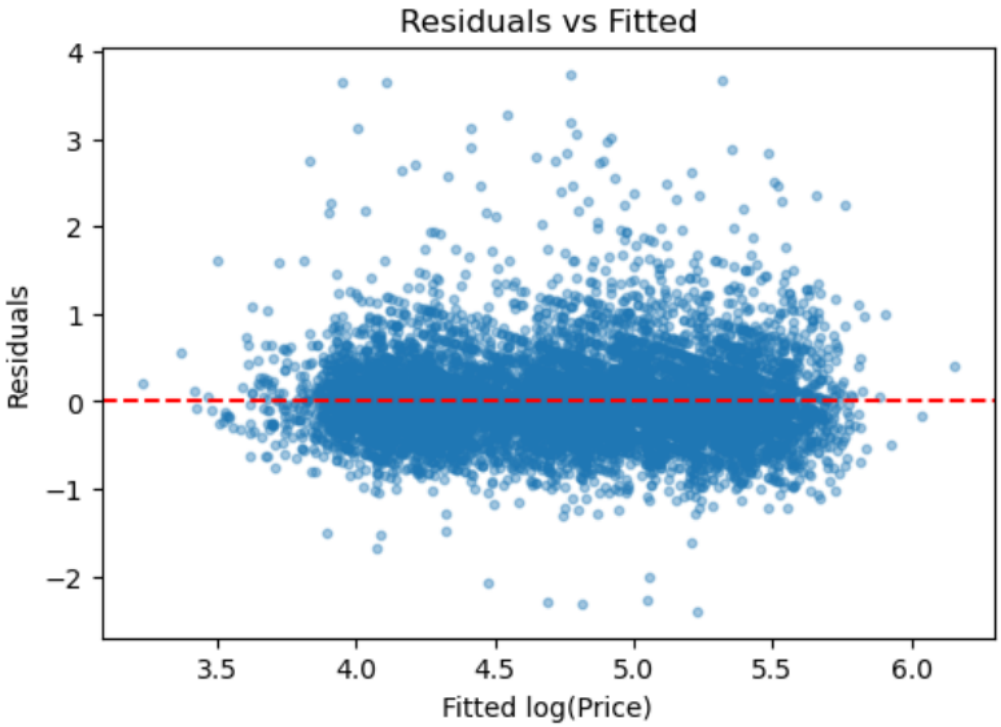


Figure B1: Residuals vs fitted values for the linear regression model.

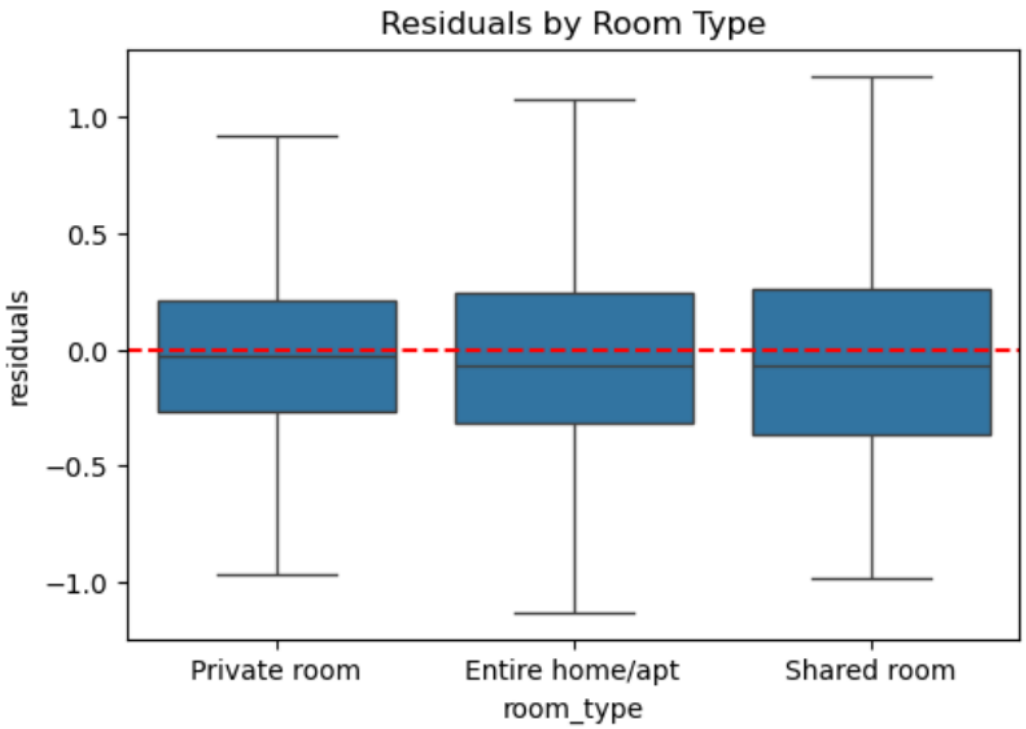


Figure B2: Distribution of regression residuals across room types.

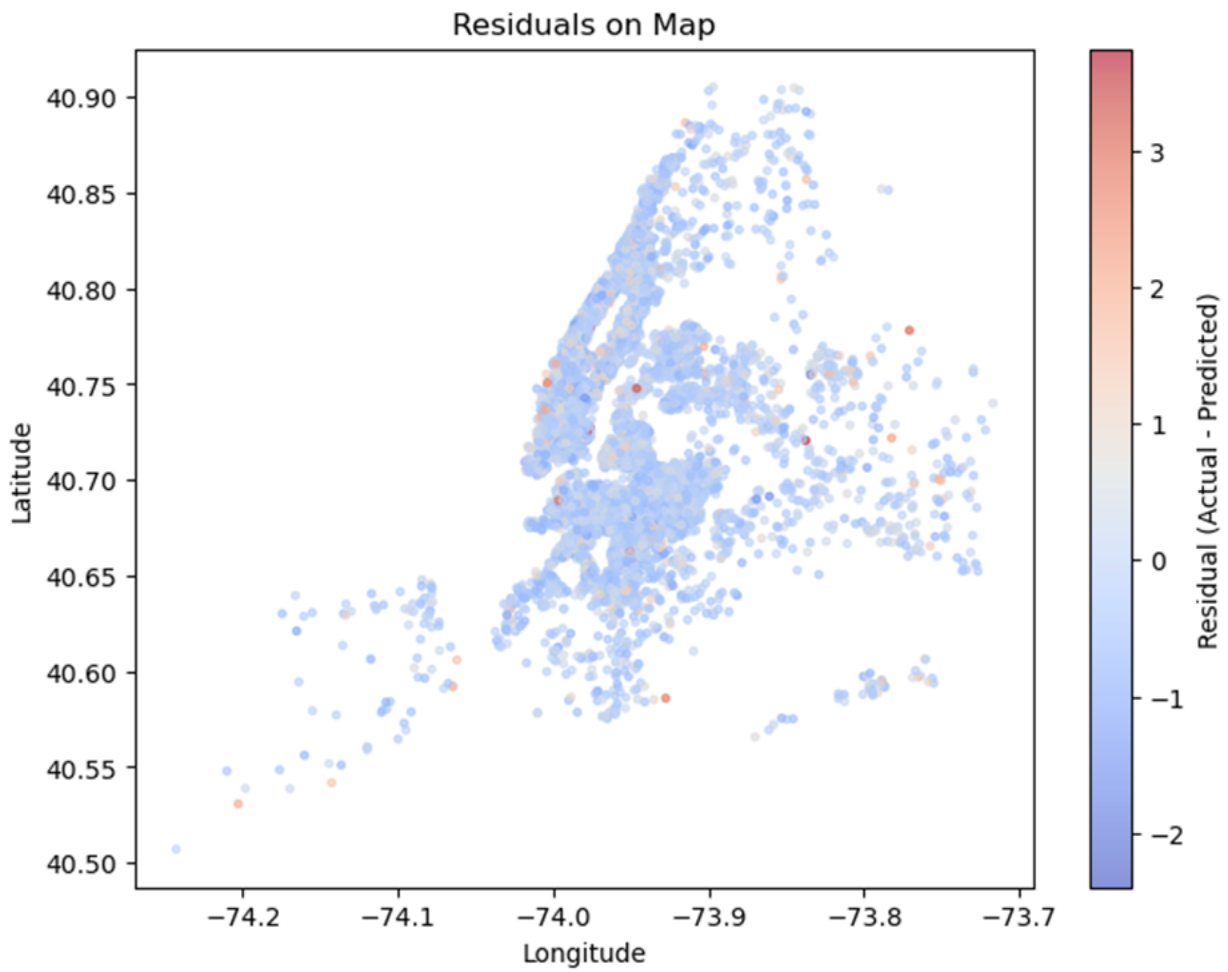


Figure B3: Geographic distribution of regression residuals across New York City listings.

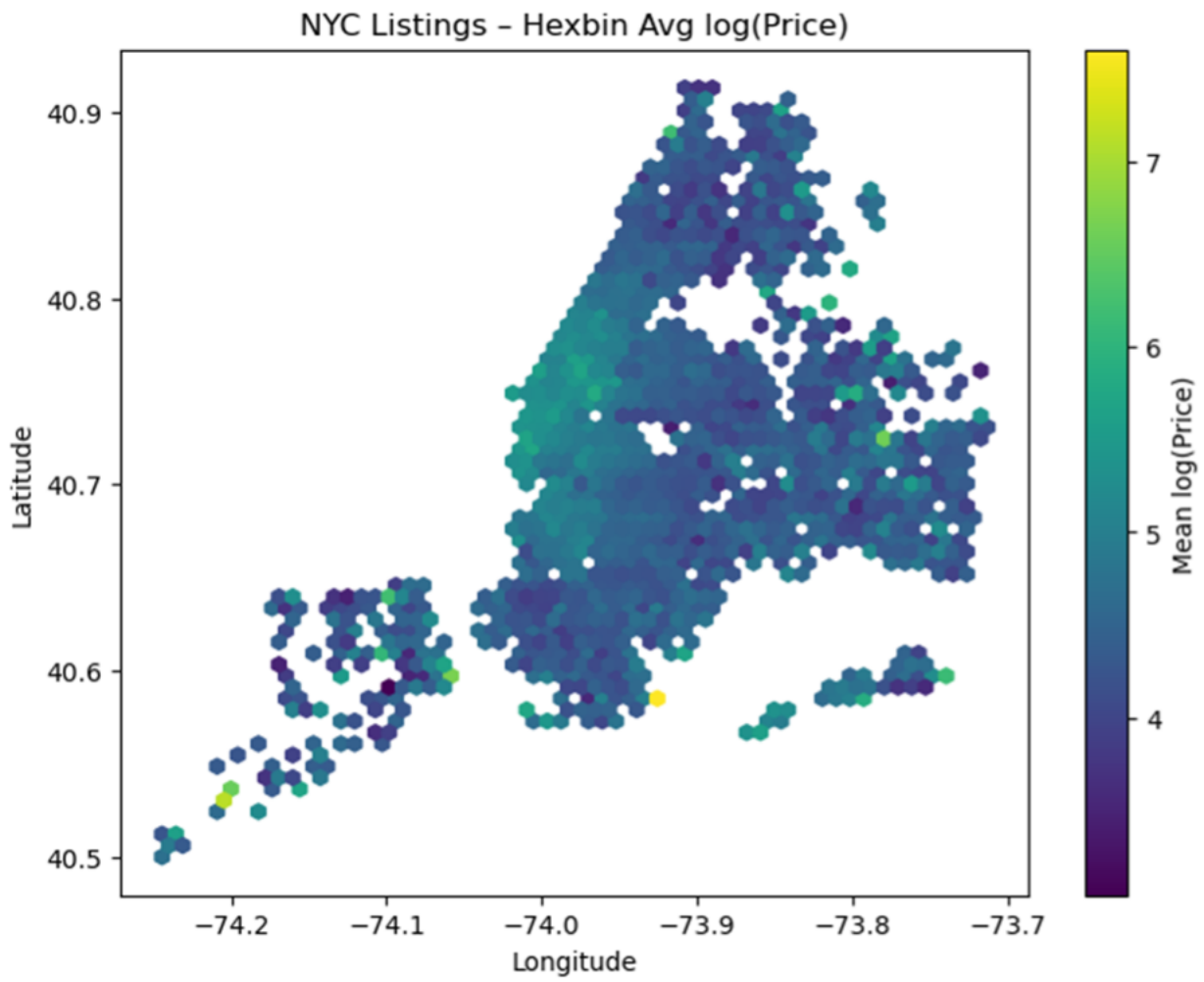


Figure B4: Spatial distribution of average log-transformed Airbnb prices across New York City (hexbin map).

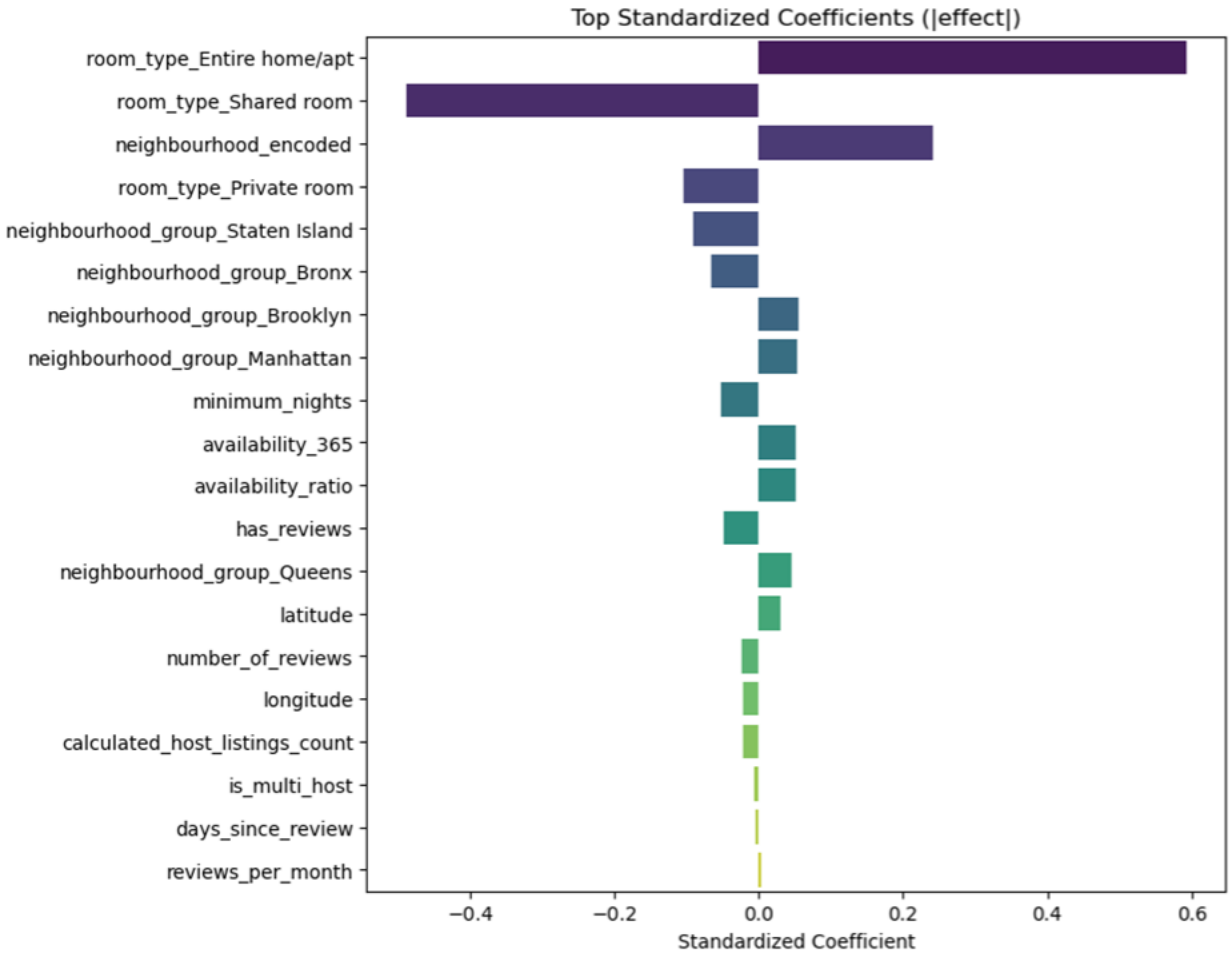


Figure B5: Standardised regression coefficients showing the relative influence of listing features on log price.

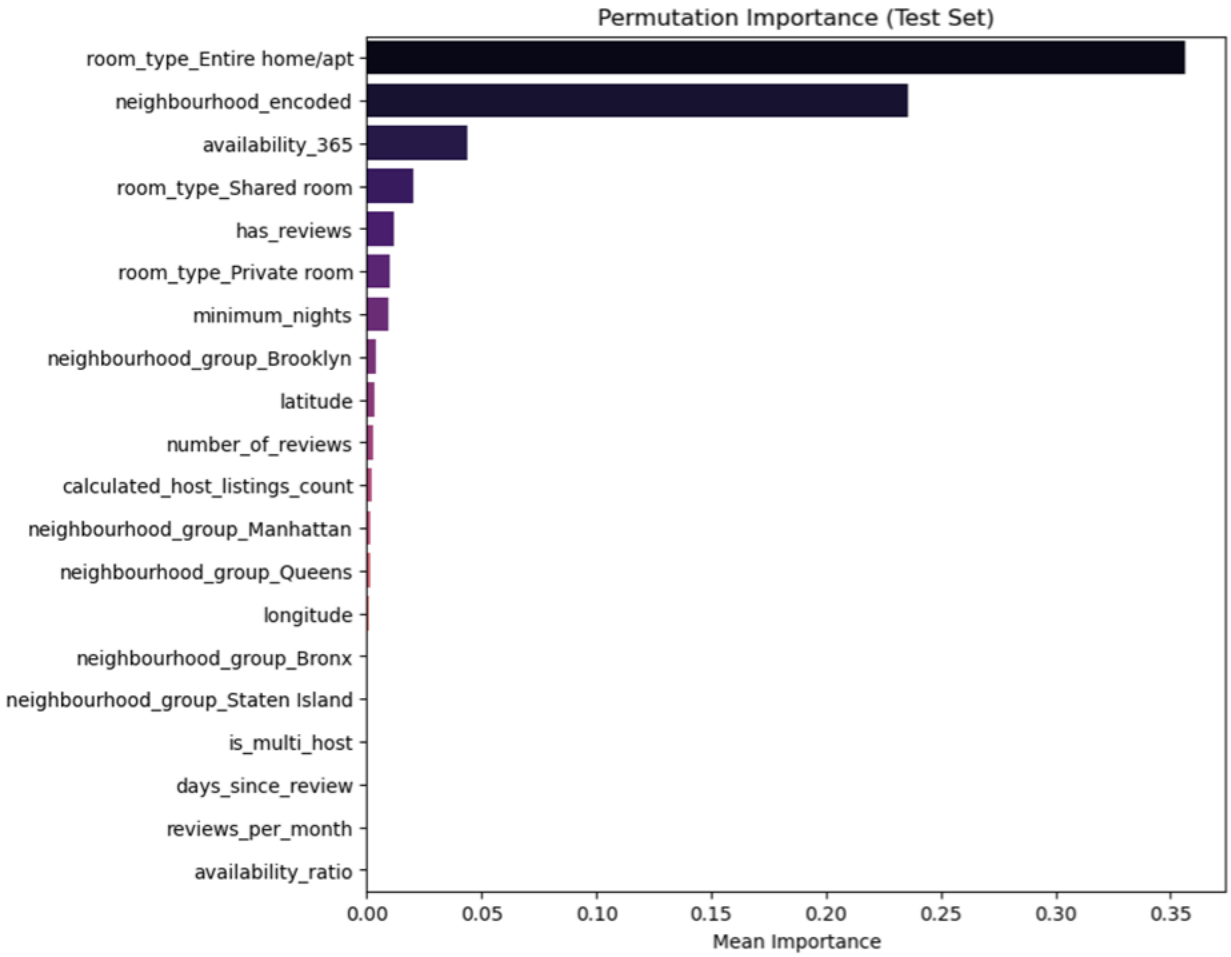


Figure B6: Permutation feature importance for the regression model based on the test dataset.

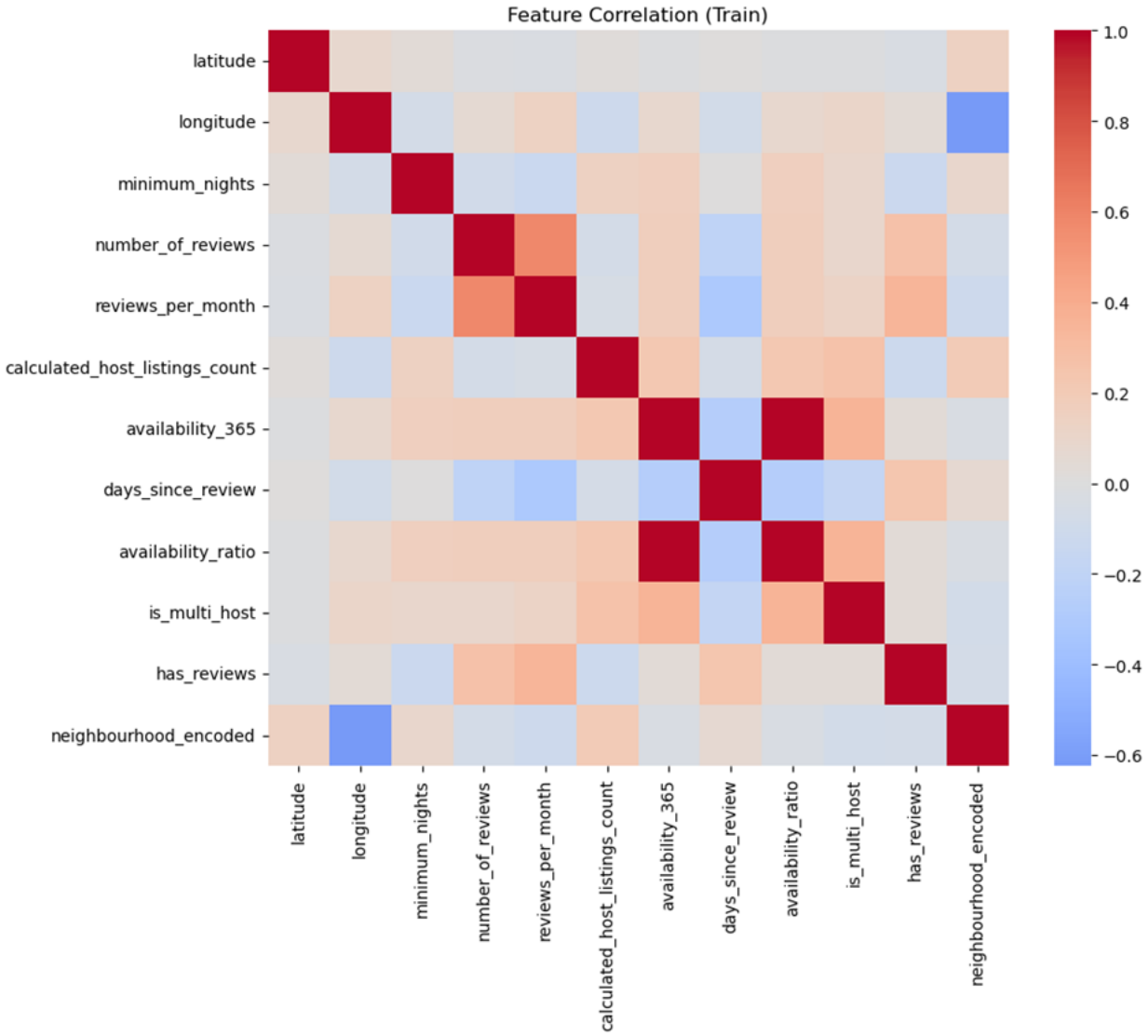


Figure B7: Correlation matrix of training features used in the regression model.

## Appendix C – Clustering Analysis

**Key Findings from K-Means Clustering**

Four distinct market segments were identified:

1. Low Activity Listings

Low review activity, high availability, and weak demand signals. These listings may represent inactive or underperforming supply.

2. Standard Listings

Moderate pricing and balanced demand characteristics. This segment represents the core market baseline.

3. Investment Listings

Higher prices and hosts managing multiple listings. These behave more like structured hospitality businesses.

4. High-Demand Value Listings

Competitive pricing with high review frequency and strong occupancy signals. These listings compete primarily on value and turnover.

KMeans clustering identifies clear groupings of listings with distinct average price levels, reflecting meaningful segmentation tied to location density and surrounding amenities. DBSCAN provides complementary insight by detecting natural density-based shapes and isolating low-density areas as noise revealing more granular spatial structures than KMeans. The K-distance graph supports DBSCAN tuning, helping identify the optimal eps threshold for neighbourhood density segmentation.

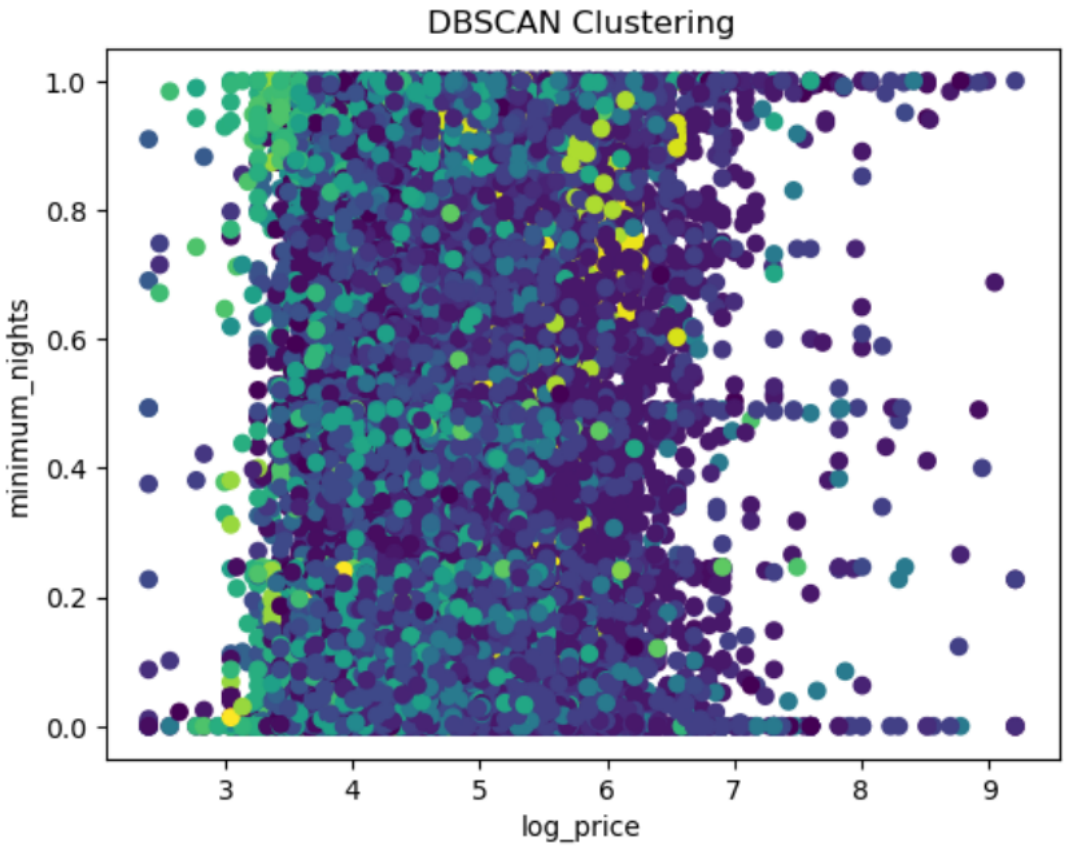


Figure C1: DBSCAN clustering results for Airbnb listings based on pricing and minimum nights.

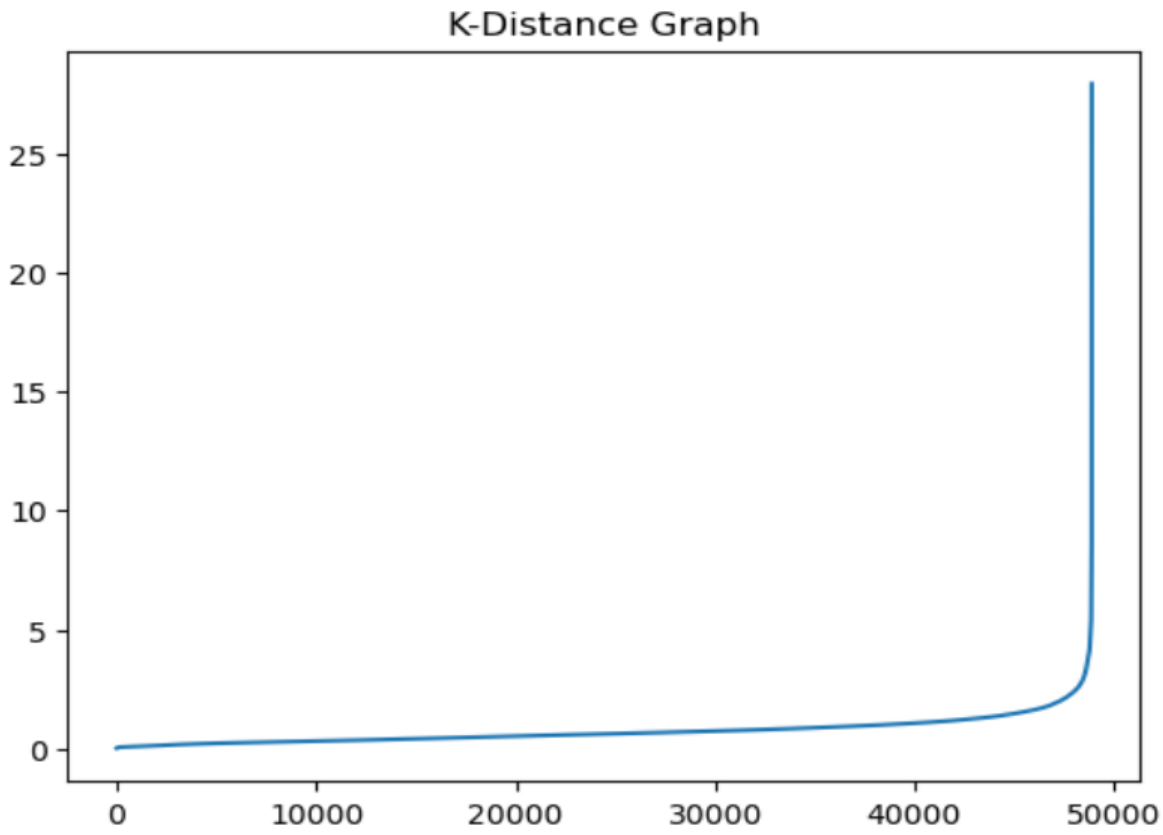


Figure C2: K-distance graph used to determine the epsilon parameter for DBSCAN clustering.